

Course – Research Methodology & Seminar (RM&S)
Mentor – Prof. Purvi Patel
Department of Electronics, ITNU

ENERGY HARVESTING POWER MANAGEMENT UNIT

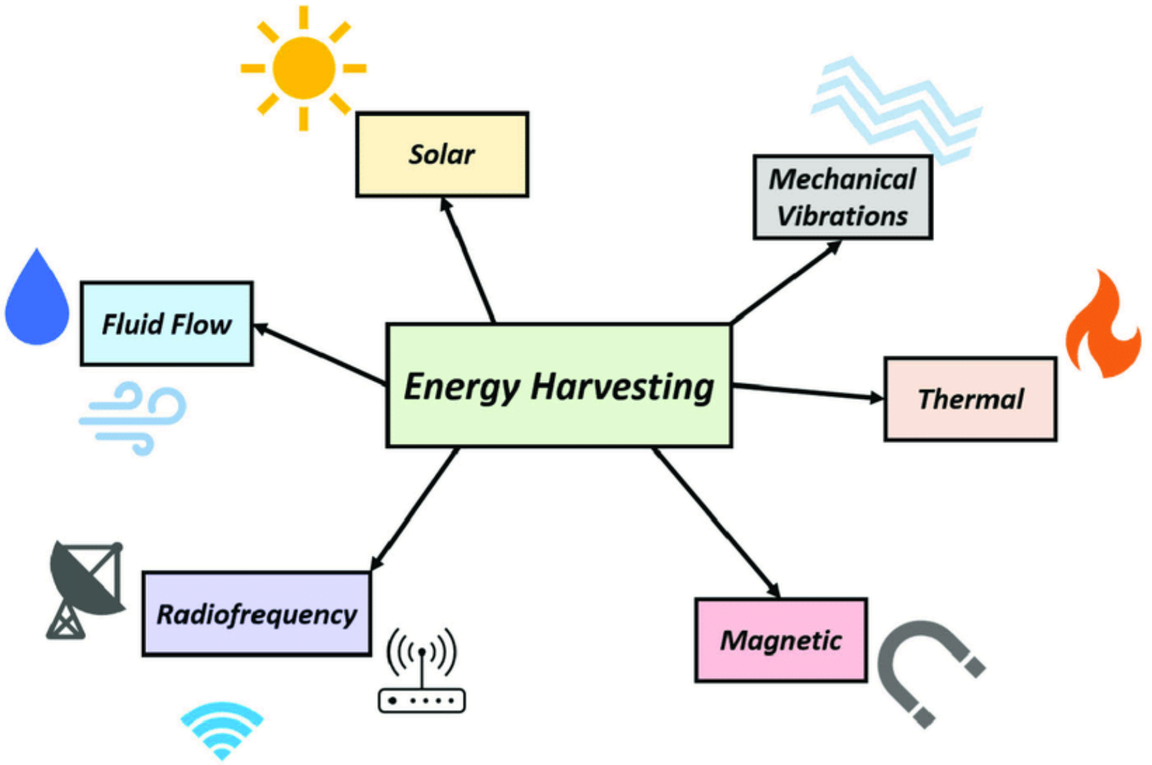
Presented by,
Purvesh Vaghela(23BEC201)
Utsav Raiyani(23BEC199)

Outline

- Introduction
- Block Diagram of PMU
- Present Work
- Conclusion
- Future Work
- References

Introduction

- The rapid advancement of Internet of Things (IoT) and Wireless Sensor Networks (WSNs) has increased the need for self-powered and energy-autonomous systems.
- Energy harvesting techniques such as solar, thermal, RF, and piezoelectric sources can generate small amounts of power from the environment.
- Among all sources, solar energy harvesting is widely preferred due to its availability and practical power output, especially for ultra-low-power applications.



Introduction

- However, these sources produce:
  - Very low voltage levels (≈0.3 V – 1 V)
  - Highly unstable and fluctuating outputs
- Due to this, harvested energy cannot be directly used to power electronic circuits.
- To address this challenge, a Power Management Unit (PMU) is required, which:
  - Regulates the output voltage
  - Improves overall energy efficiency
  - Enables reliable and continuous operation of the system

Objective

To design an Energy Harvester-powered, Power Management Unit for Low Power Applications using 0.18um CMOS in Cadence Virtuoso.

Key Takeways

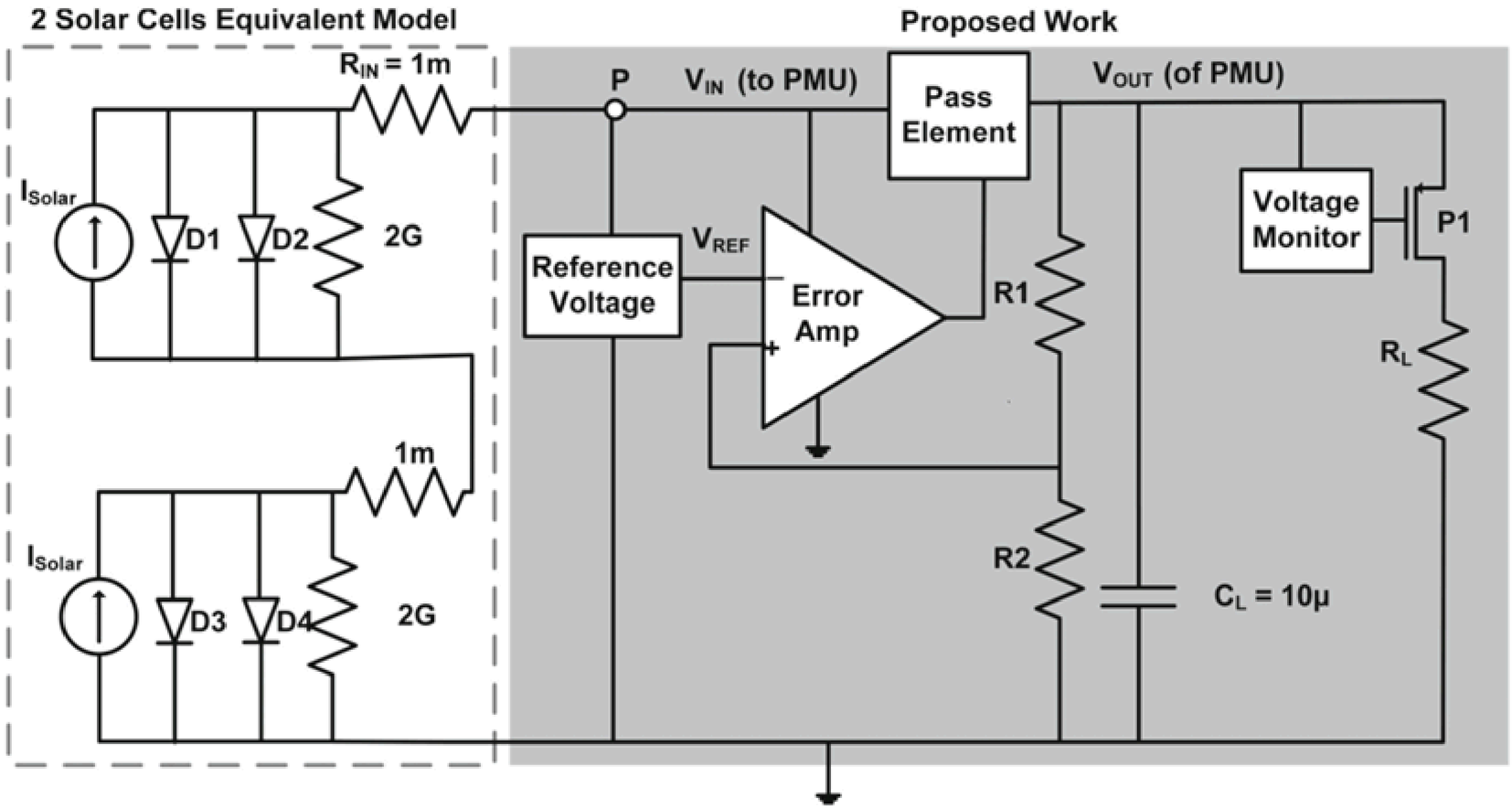
Solar Harvester

- Higher ambient power density
- Suitable voltage for CMOS circuits
- Reliable energy source for IoT nodes

LDO Regulator

- Stable low-noise output voltage
- Simple architecture for low-power systems
- Suitable for low-current sensor loads

Proposed Block Diagram of PMU



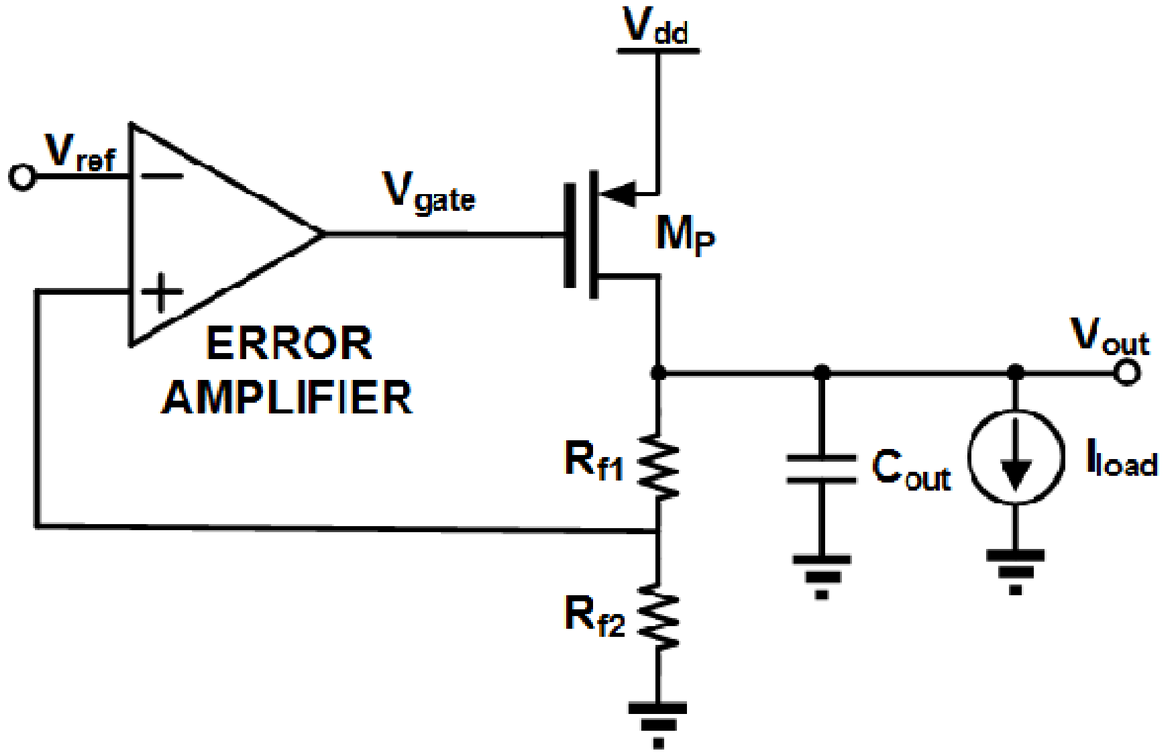
Present Work

LDO and it’s Working :

A Low Dropout (LDO) regulator is a DC linear voltage regulator circuit that provides a stable output voltage from a higher input voltage, even when the input-output differential is very small.

Key Components:

- Error amplifier
- Pass transistor
- Feedback network



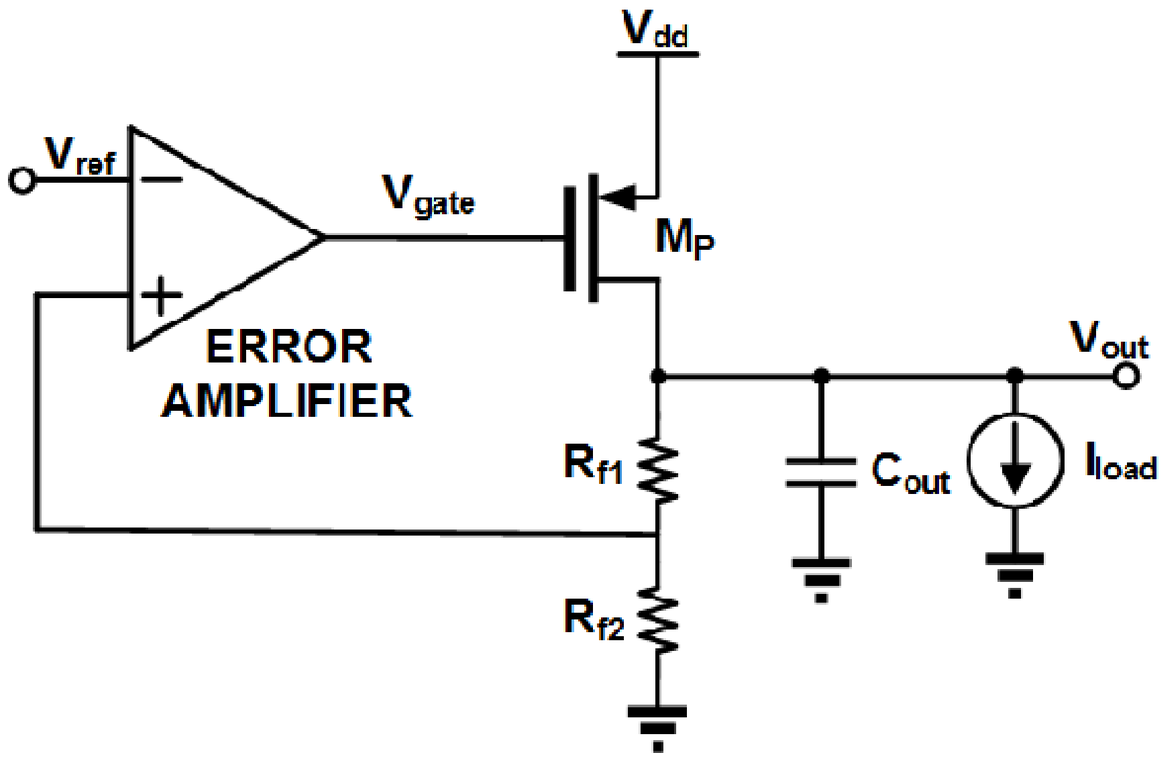
Source: https://www.researchgate.net/figure/Conventional-LDO-block-diagram_fig1_318473208

Present Work

LDO and it’s Working :

Working Principle:

- Output voltage is sensed using resistor divider
- Compared with reference voltage (Vref)
- Error amplifier generates control signal
- Pass transistor adjusts current flow



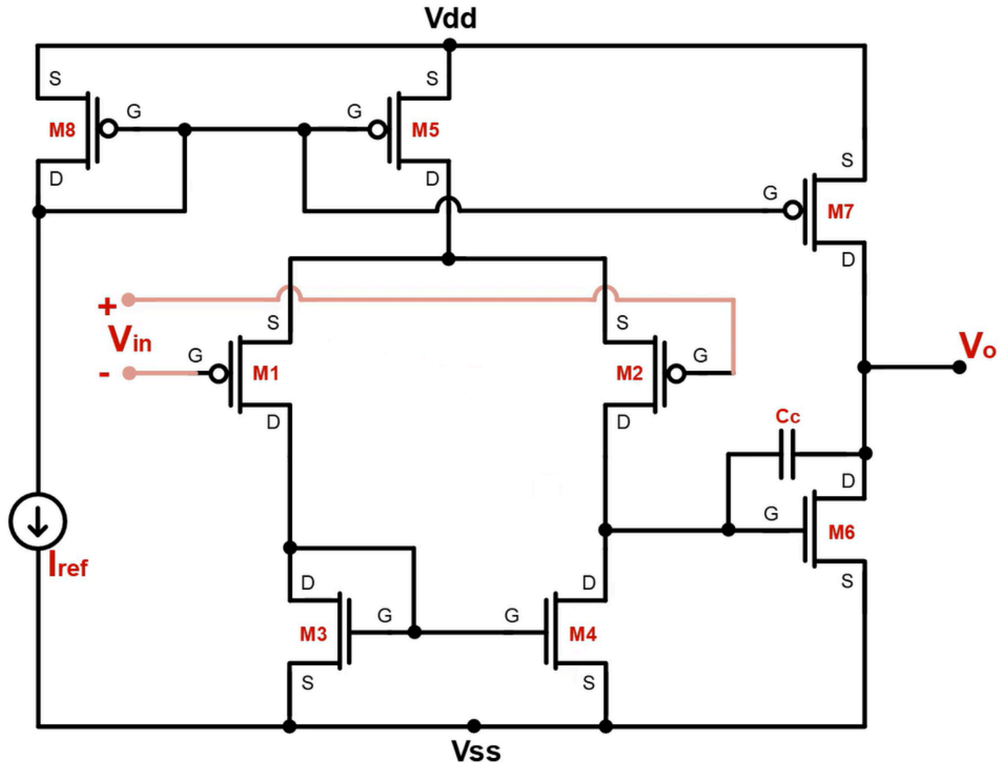
“LDO acts like a feedback-controlled system to maintain constant output voltage.”

Error Amplifier :

An error amplifier is a high-gain differential amplifier used in power management circuits to compare a sampled output voltage against a precise reference voltage and amplify the difference (error) to stabilize the system

It compares:

- Reference Voltage (Vref)
- Feedback Voltage (Vfb) (from output)
- It generates an error signal = (Vref – Vfb)
- This signal is used to control the pass transistor



Source: https://pubs.aip.org/aip/acp/article-pdf/doi/10.1063/1.5142132/14205631/020040_1_online.pdf

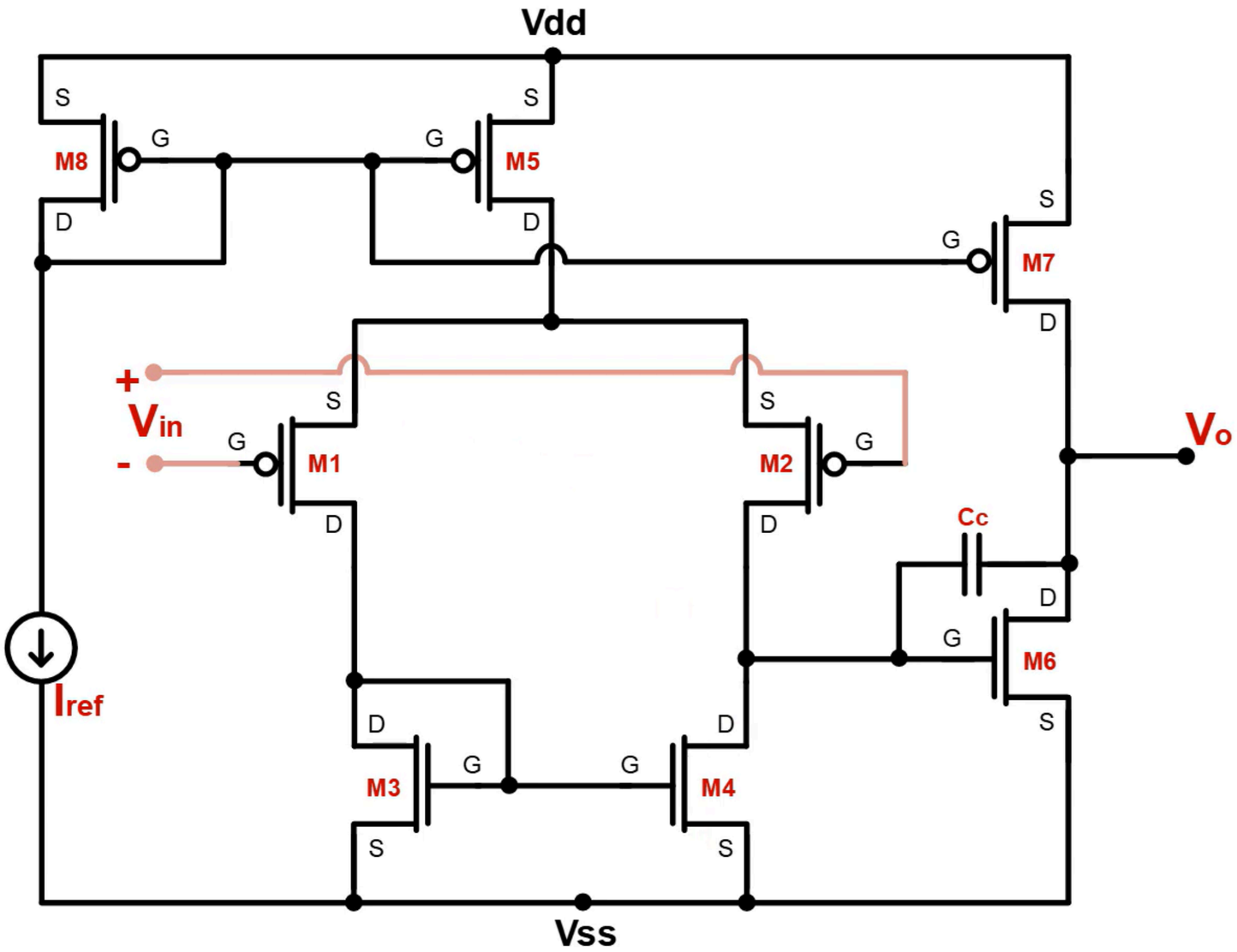
Design Specifications for ECG(Electrocardiogram) & EMG(Electromyography)

Parameter / Specification	Target Value
Common-Mode Rejection	> 80 - 100 dB
Power Dissipation	< 10 - 100 uW
Supply Voltage	1.0 - 1.8 V
DC Open-Loop Gain	> 60 - 80 dB
Gain Bandwidth Product	10 - 2000 kHz
Phase Margin	> 60 deg
Slew Rate	> 0.1 - 1 V/us

Architectural & Topology Selection

- **Input Pair Sizing Strategy:** PMOS input differential pair — selected because V_ICM (≈0.5 V) is too low for an NMOS pair to sense, and PMOS gives lower 1/f noise for biomedical signals.
- **Biasing Region:** Strong/moderate inversion (square-law), with all transistors kept in saturation — selected because 1.8 V gives enough headroom for predictable square-law sizing, and saturation is required for high output impedance (gain) and accurate current mirroring.
- **Compensation Topology:** Miller compensation with nulling resistor Rz — selected because Rz cancels the right-half-plane zero that would otherwise erode phase margin.

Architecture :



Calculations :

Master formula

$$\frac{W}{L} = \frac{2 \cdot I_D}{k \cdot V_{ov}^2}$$

Constants: k_n = 316 µA/V² (NMOS), k_p = 62.5 µA/V² (PMOS)

M8 — Bias diode (PMOS)

I_D = 1 µA (defined — this is I_ref itself) · V_ov = 0.20 V (chosen: current-source device, prioritize headroom)

$$\frac{W}{L} = \frac{2(1\mu)}{62.5\mu \times 0.20^2} = \frac{2\mu}{2.5\mu} = 0.80$$

→ **W = 0.80 µm, L = 1 µm**

M5 — Tail current source (PMOS)

I_D = 1 µA (mirrors M8, $W_5/W_8 = 1$) · V_ov = 0.20 V

$$\frac{W}{L} = \frac{2(1\mu)}{62.5\mu \times 0.04} = 0.80$$

→ **W = 0.80 µm, L = 1 µm**

Calculations :

M1, M2 — Input pair (PMOS)

I_D = 0.5 μA (= I_tail/2 = 1μA/2, differential pair splits equally) · V_ov = 0.12 V (chosen: minimize for max g_m/I_D)

$$\frac{W}{L} = \frac{2(0.5\mu)}{62.5\mu \times 0.12^2} = \frac{1\mu}{0.9\mu} = 1.11$$

→ **W = 1.1 μm, L = 1 μm** (each)

M3, M4 — Mirror load (NMOS)

I_D = 0.5 μA (M3 carries M1's current directly; M4 mirrors M3, $W_4/W_3 = 1$) · V_ov = 0.15 V

$$\frac{W}{L} = \frac{2(0.5\mu)}{316\mu \times 0.15^2} = \frac{1\mu}{7.11\mu} = 0.14$$

→ **W = 0.4 μm (min-width bound), L = 2.855 μm** (each)

Calculations :

M6 — 2nd-stage common-source (NMOS)

I_D = 2 μA (set by KCL — must sink whatever M7 sources at OUT) · V_ov = 0.15 V (chosen: match M3/M4 for offset cancellation)

$$\frac{W}{L} = \frac{2(2\mu)}{316\mu \times 0.15^2} = \frac{4\mu}{7.11\mu} = 0.56$$

→ **W = 0.56 μm, L = 1 μm**

M7 — 2nd-stage load (PMOS)

I_D = 2 μA (mirrors M8, $W_7/W_8 = 2$) · V_ov = 0.20 V

$$\frac{W}{L} = \frac{2(2\mu)}{62.5\mu \times 0.04} = \frac{4\mu}{2.5\mu} = 1.60$$

→ **W = 1.6 μm, L = 1 μm**

Calculations :

M6 — 2nd-stage common-source (NMOS)

I_D = 2 μA (set by KCL — must sink whatever M7 sources at OUT) · V_ov = 0.15 V (chosen: match M3/M4 for offset cancellation)

$$\frac{W}{L} = \frac{2(2\mu)}{316\mu \times 0.15^2} = \frac{4\mu}{7.11\mu} = 0.56$$

→ **W = 0.56 μm, L = 1 μm**

M7 — 2nd-stage load (PMOS)

I_D = 2 μA (mirrors M8, $W_7/W_8 = 2$) · V_ov = 0.20 V

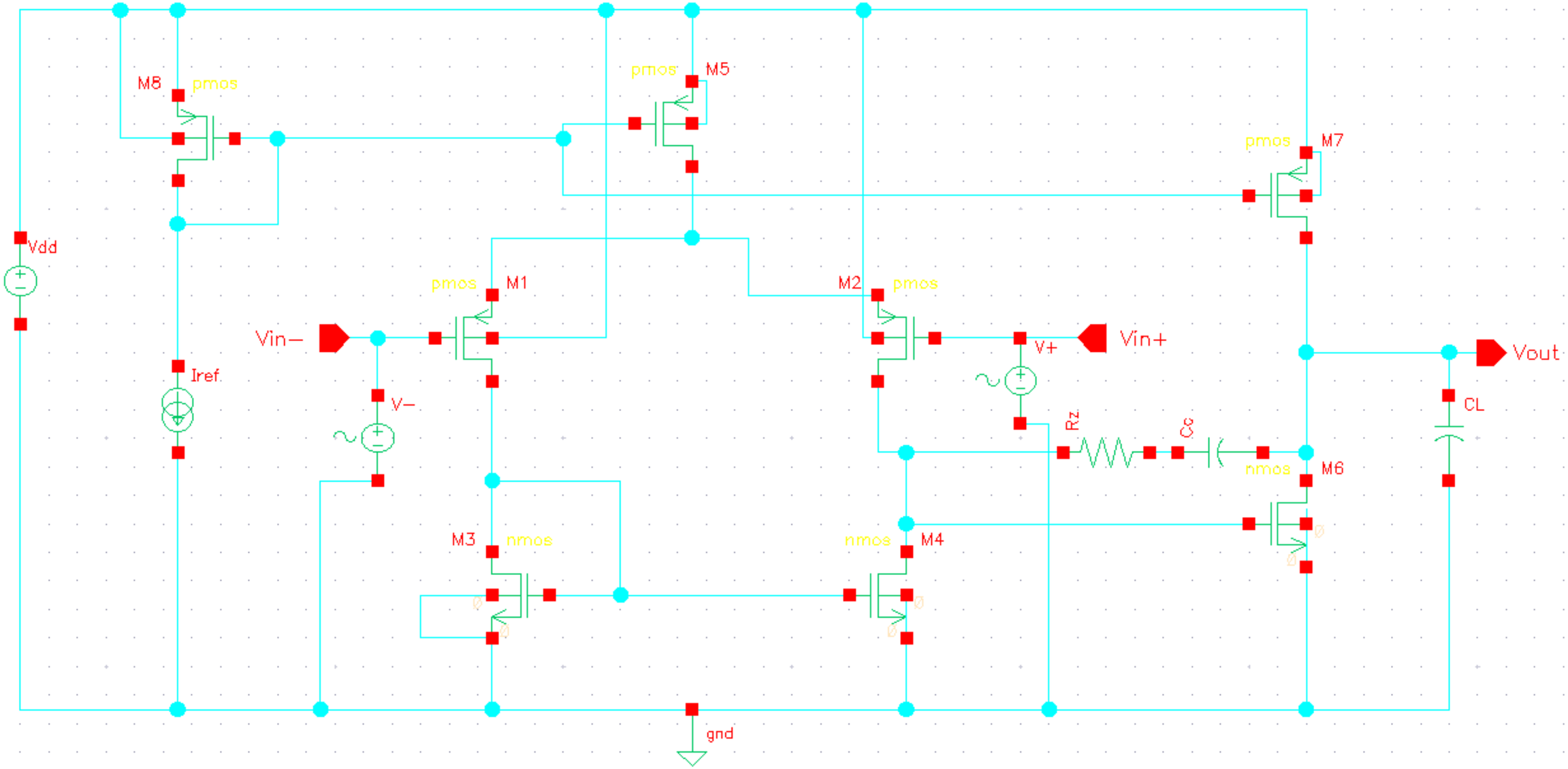
$$\frac{W}{L} = \frac{2(2\mu)}{62.5\mu \times 0.04} = \frac{4\mu}{2.5\mu} = 1.60$$

→ **W = 1.6 μm, L = 1 μm**

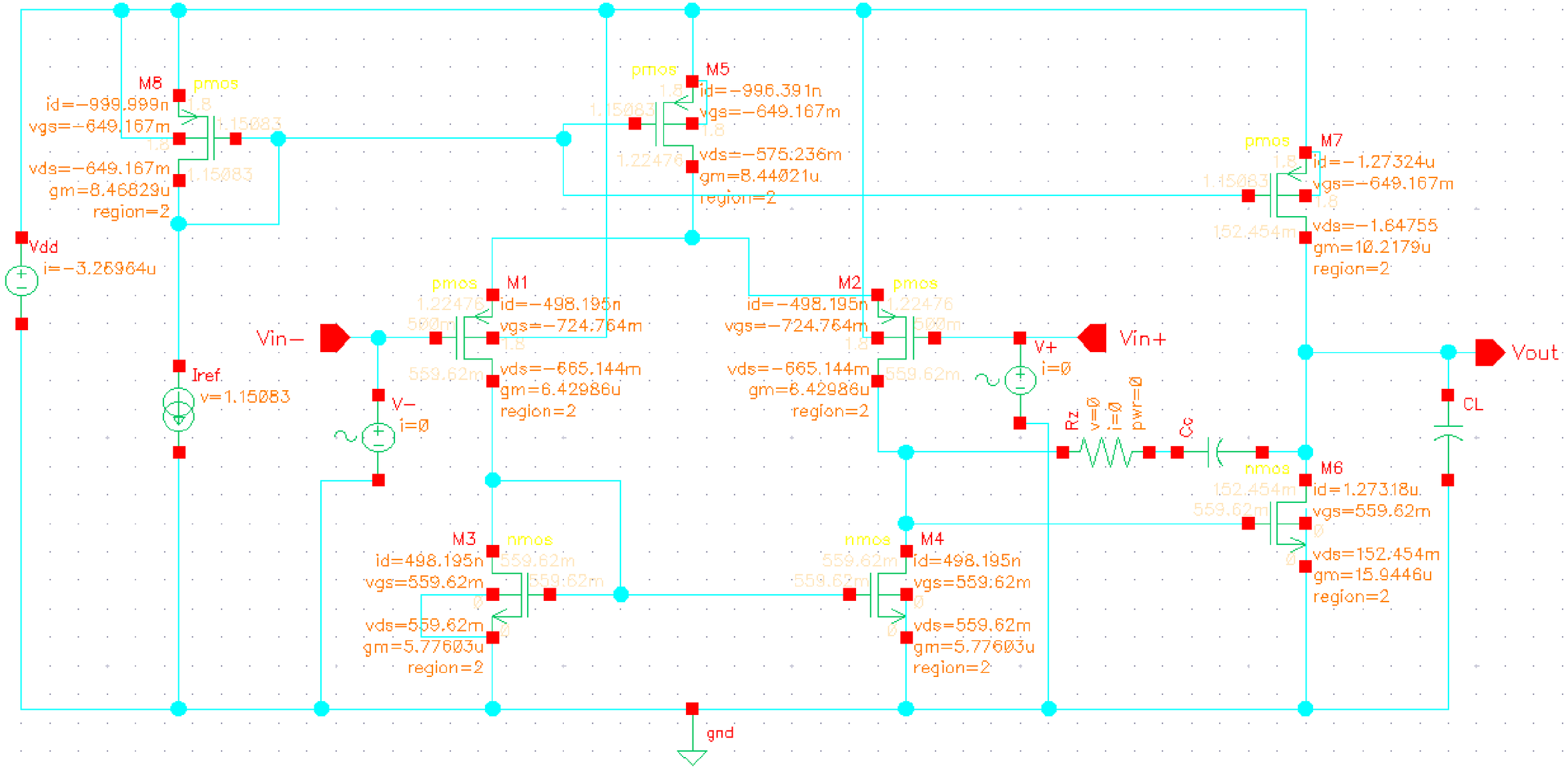
Calculated Table :

Device	I_D	V_ov	W/L (calc)	W	L	Region
M1, M2	0.5 µA	0.12 V	1.11	1.1 µm	1 µm	2
M3, M4	0.5 µA	0.15 V	0.14	0.4 µm	2.855 µm	2
M5	1.0 µA	0.20 V	0.8	0.8 µm	1 µm	2
M6	2.0 µA	0.15 V	0.56	0.56 µm	1 µm	2
M7	2.0 µA	0.20 V	1.6	1.6 µm	1 µm	2
M8	1.0 µA	0.20 V	0.8	0.8 µm	1 µm	2

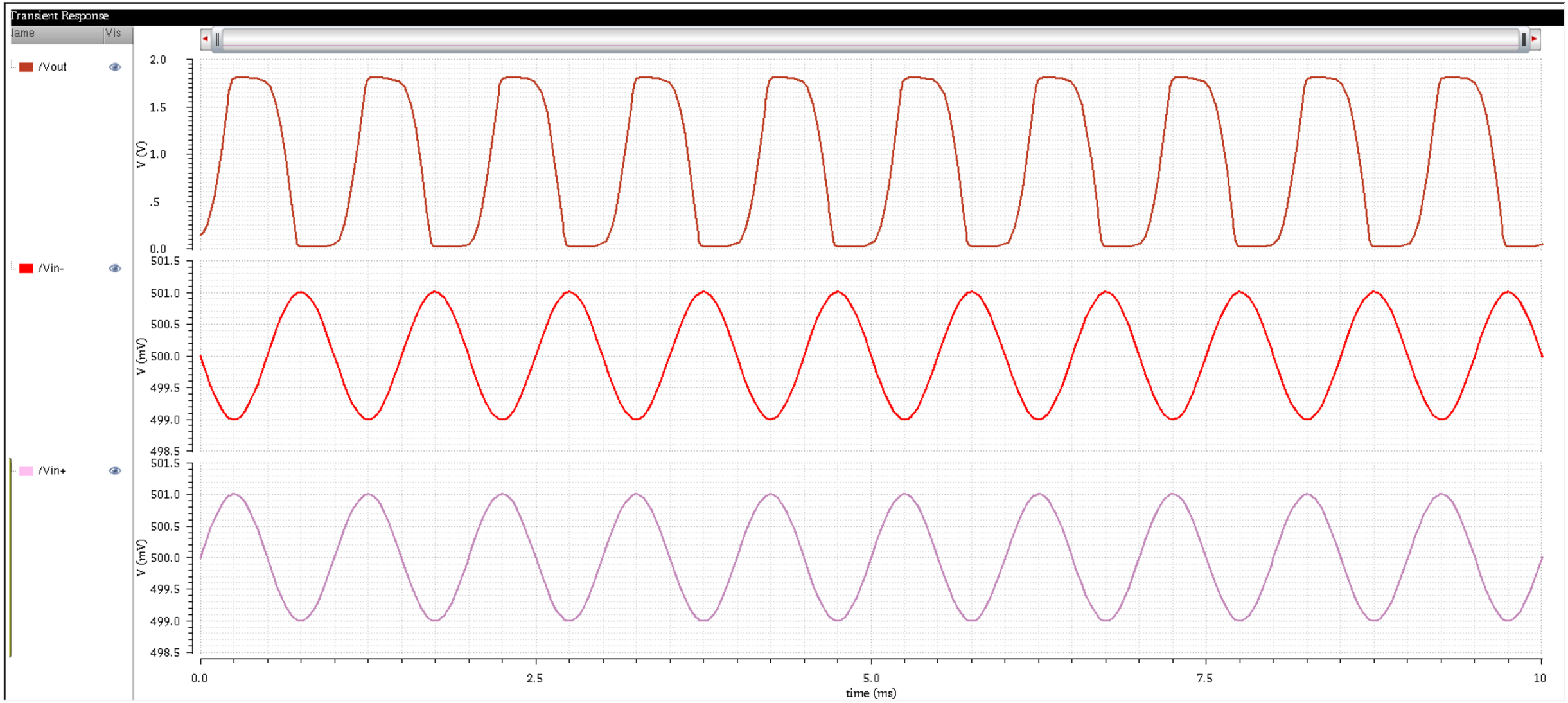
Schematic :


M8
pmos
M5
pmos
M7
pmos
Vdd
M1
pmos
M2
pmos
Vin−
Vin+
V+
Iref
V−
Rz
Cc
nmos
M6
CL
Vout
M3
nmos
nmos
M4
gnd

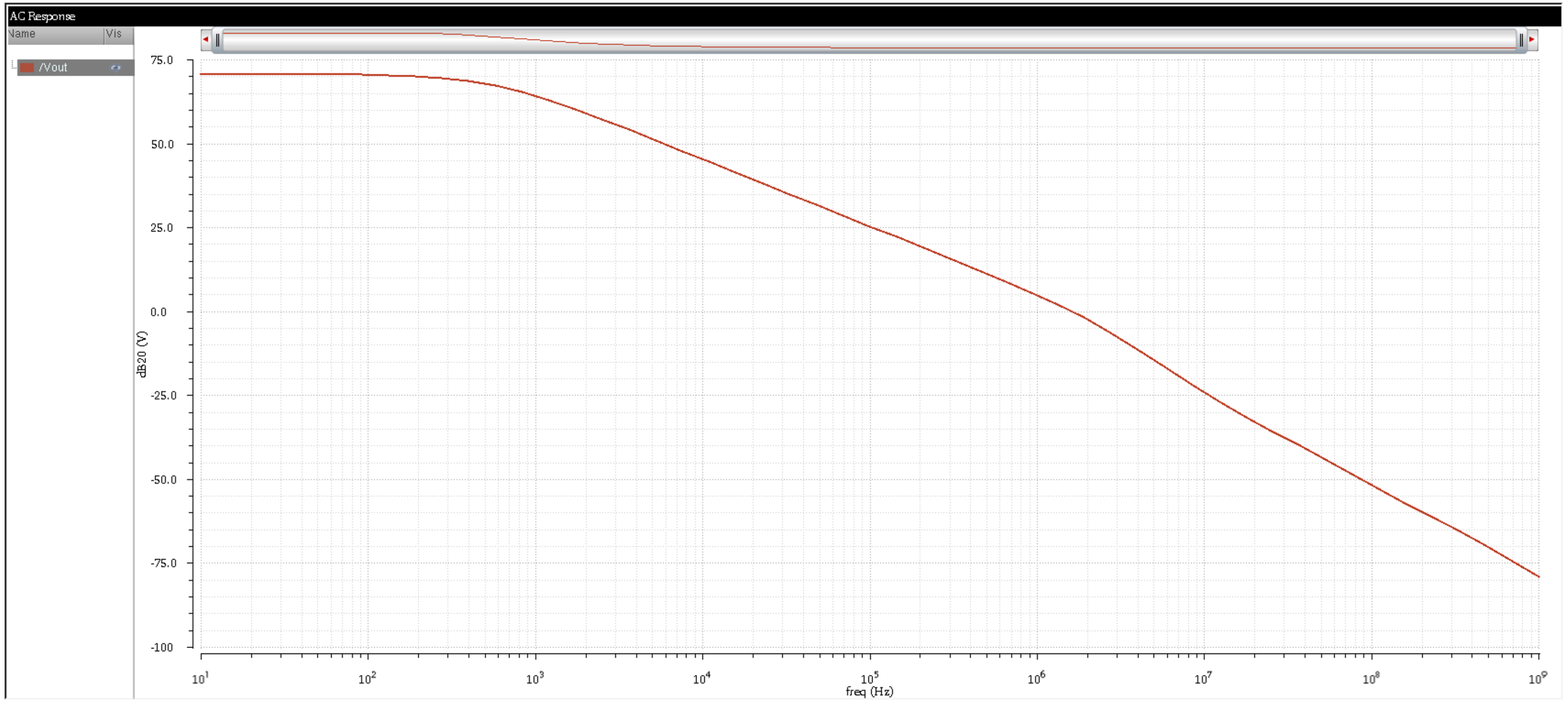
DC dummy Analysis :



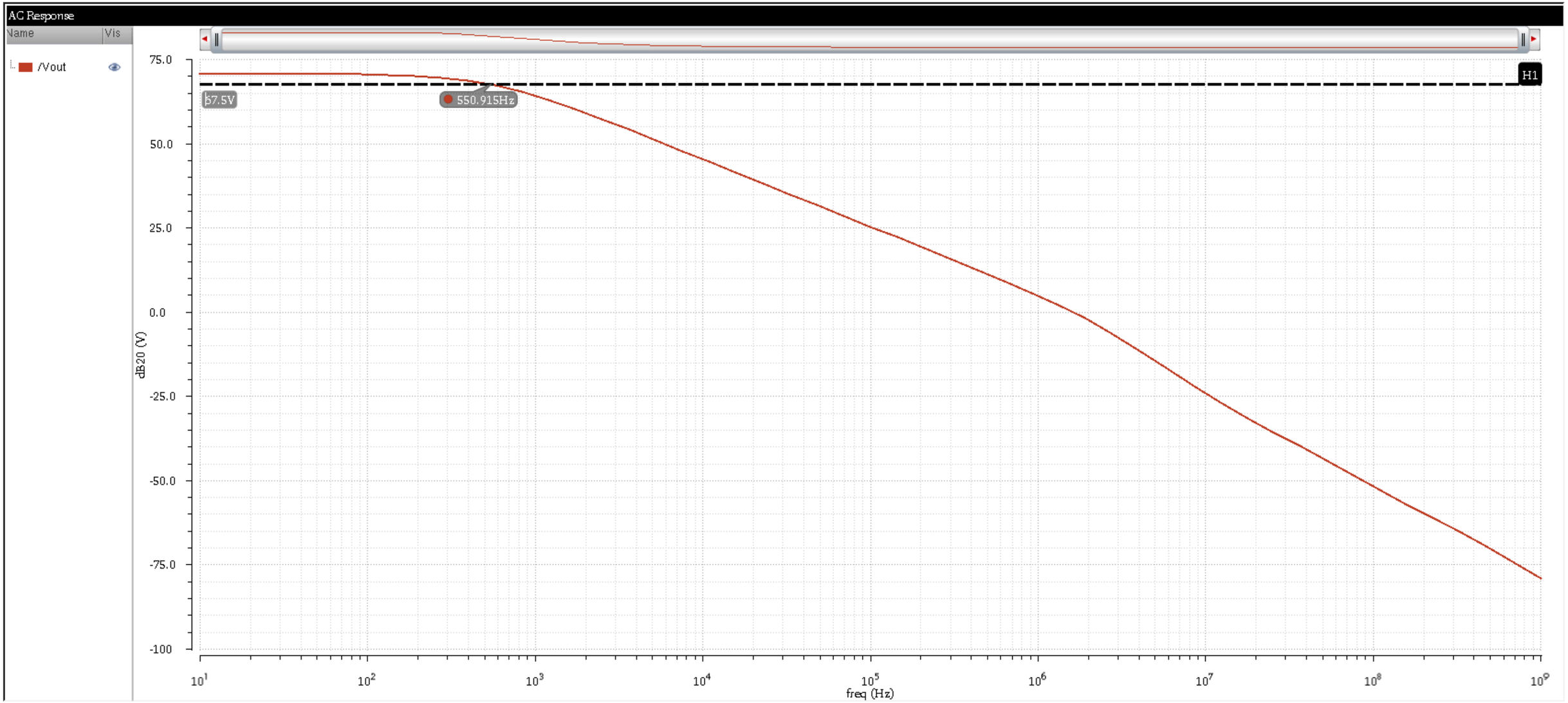
Transient Response :



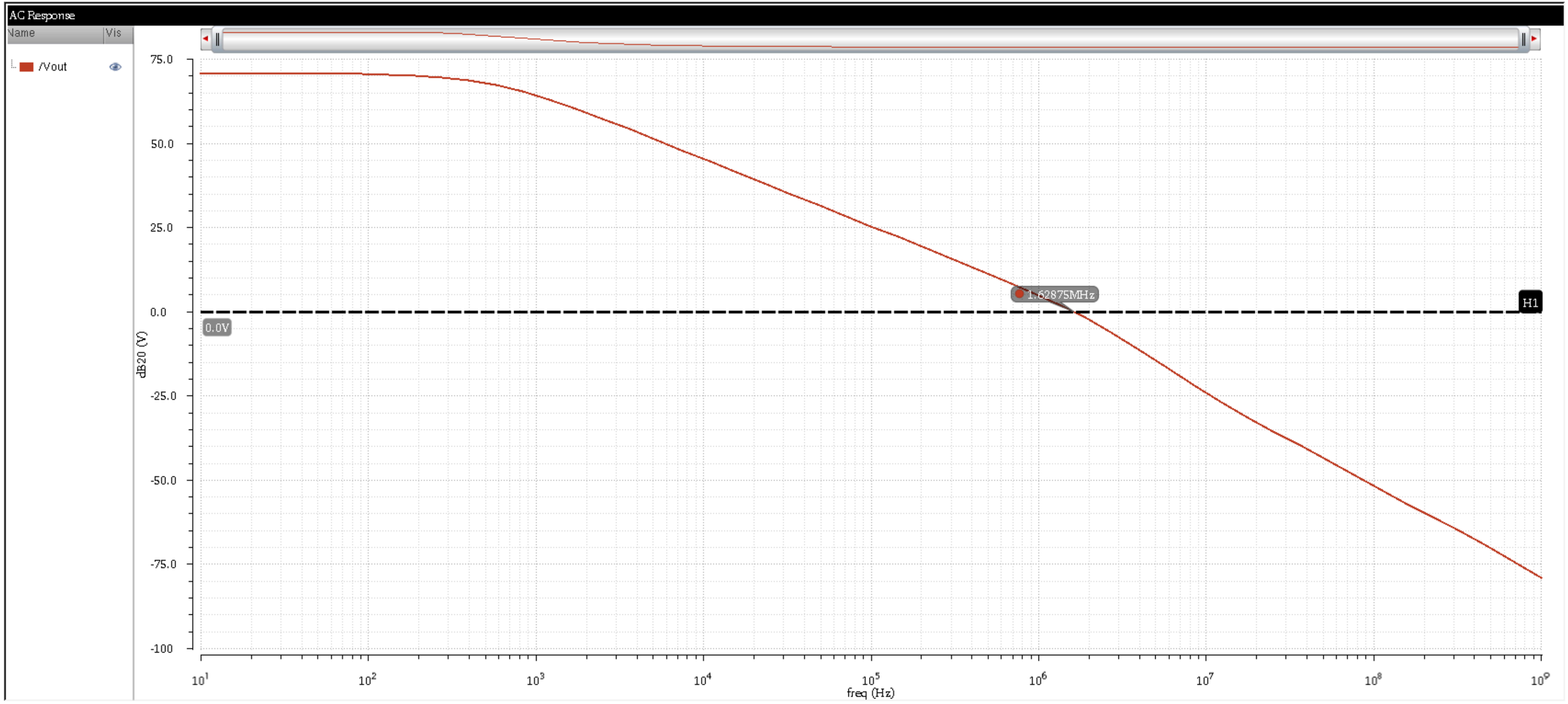
Frequency vs Gain response :



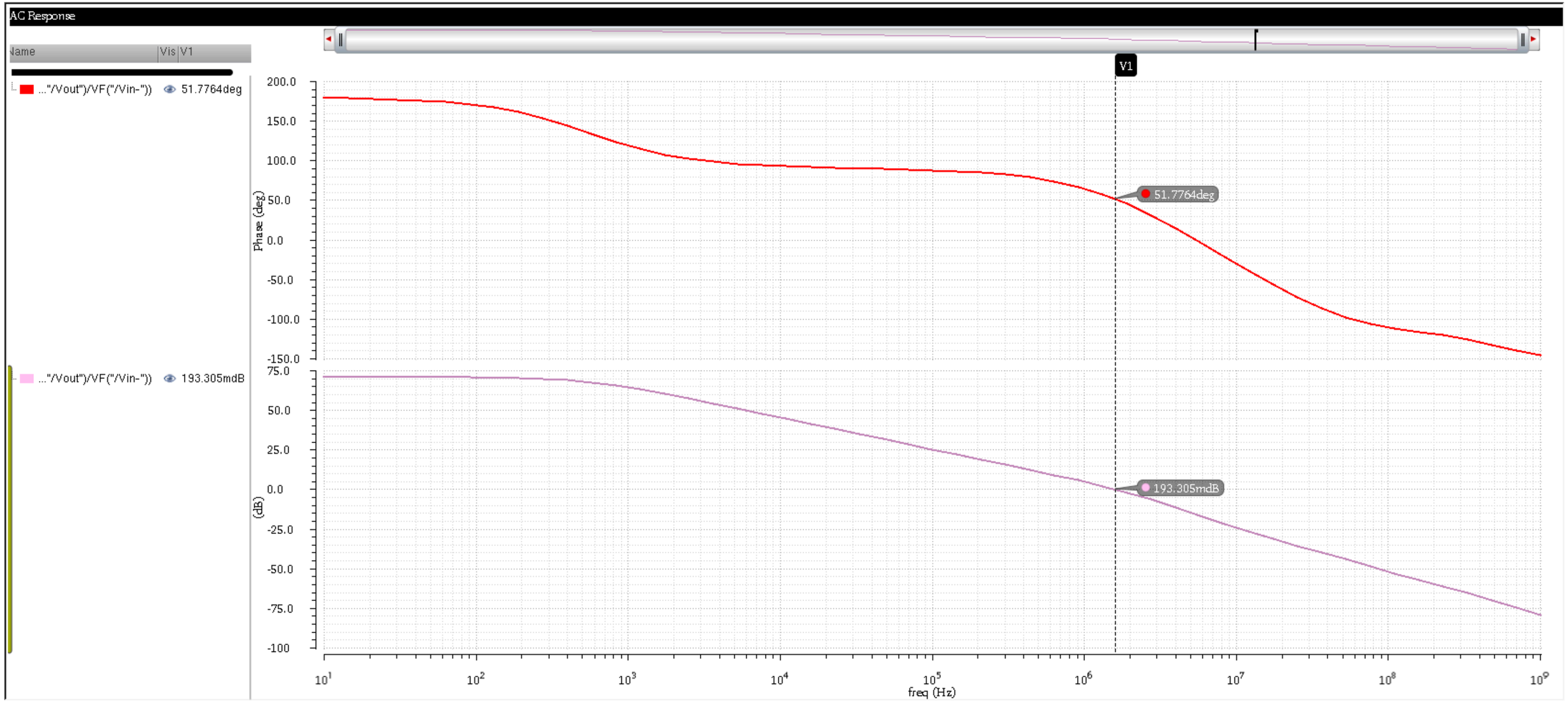
-3db Gain :


AC Response
Name
Vis
/Vout
75.0
50.0
25.0
0.0
-25.0
-50.0
-75.0
-100
dB20 (V)
H1
37.5V
550.915Hz
10^1
10^2
10^3
10^4
10^5
10^6
10^7
10^8
10^9
freq (Hz)

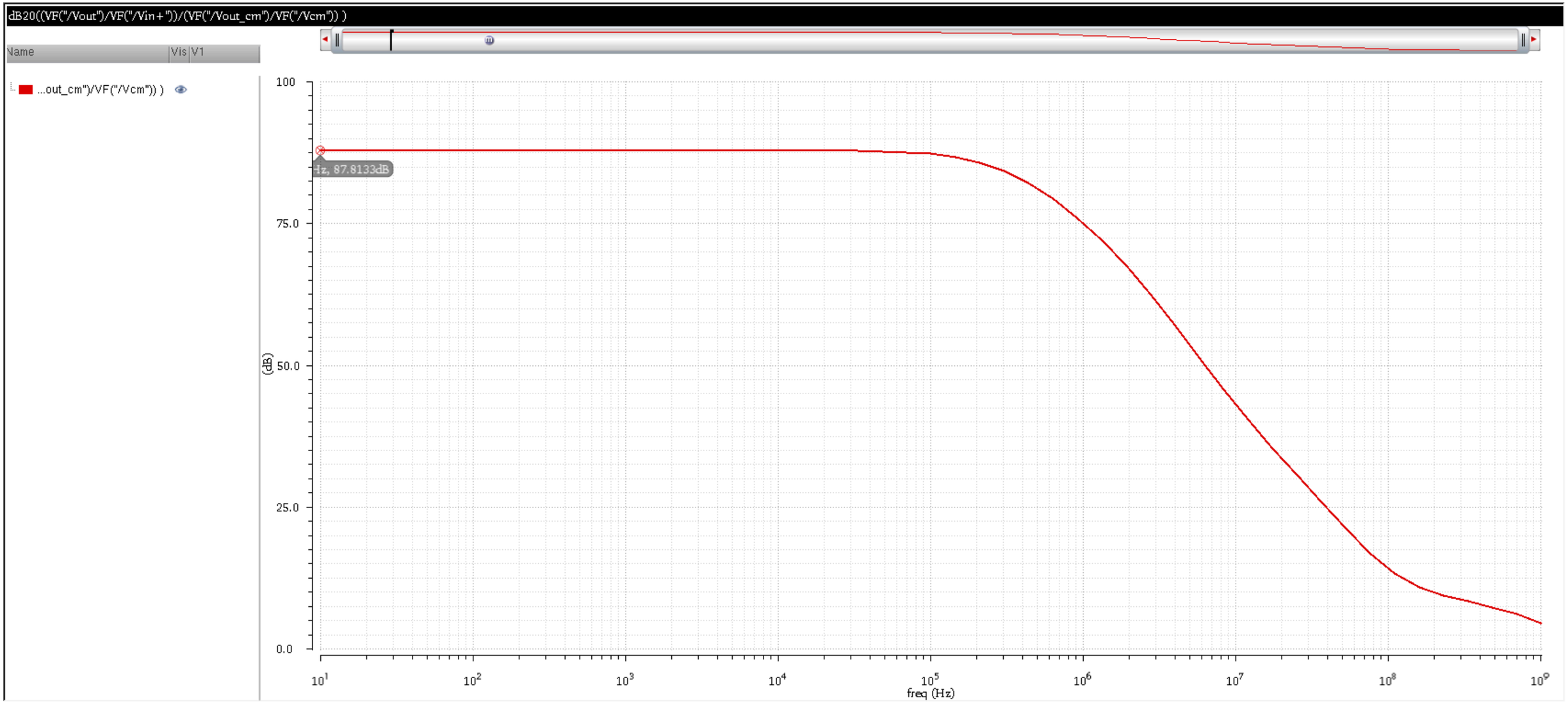
Unity Gain Bandwidth :



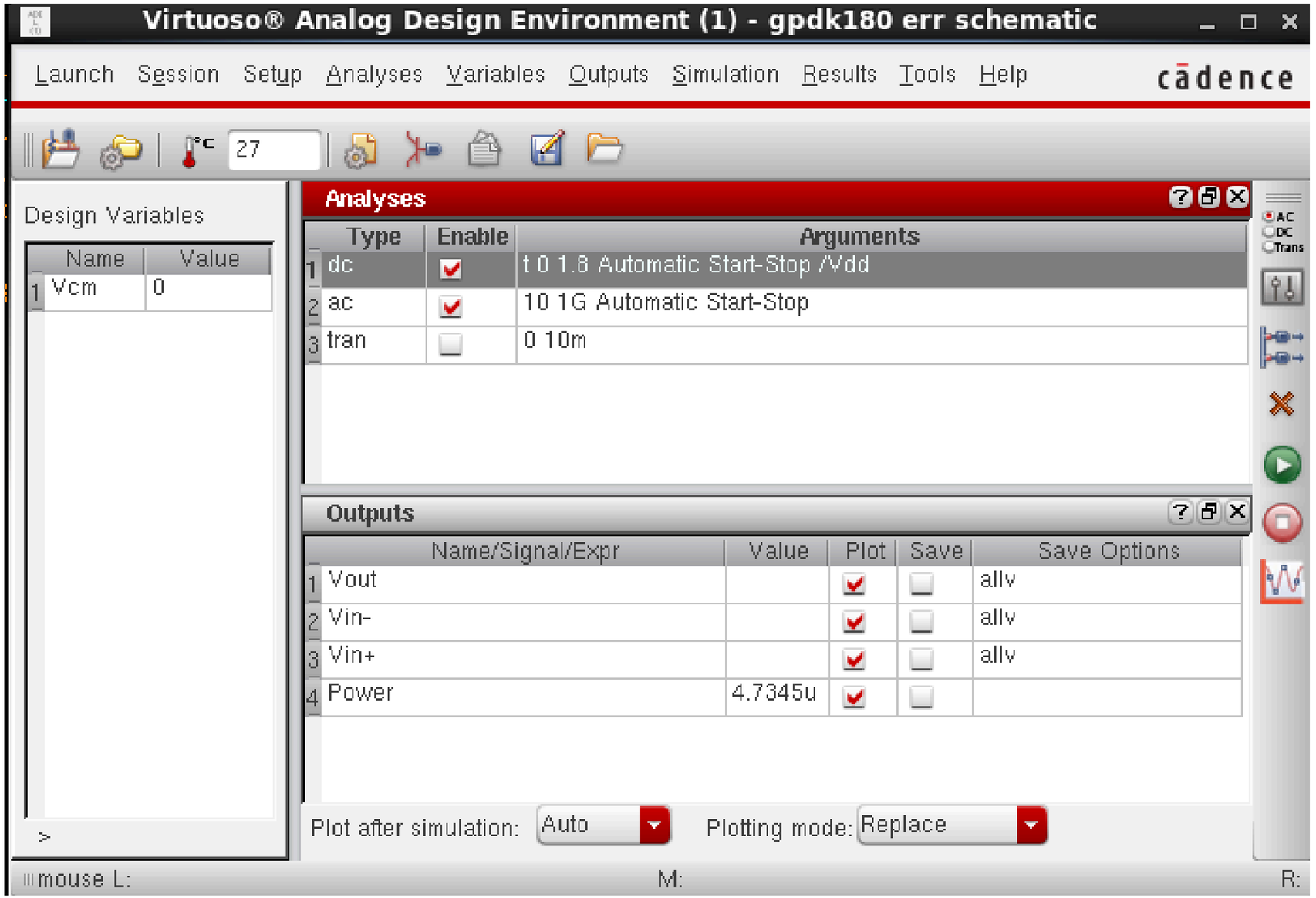
Phase Margin :



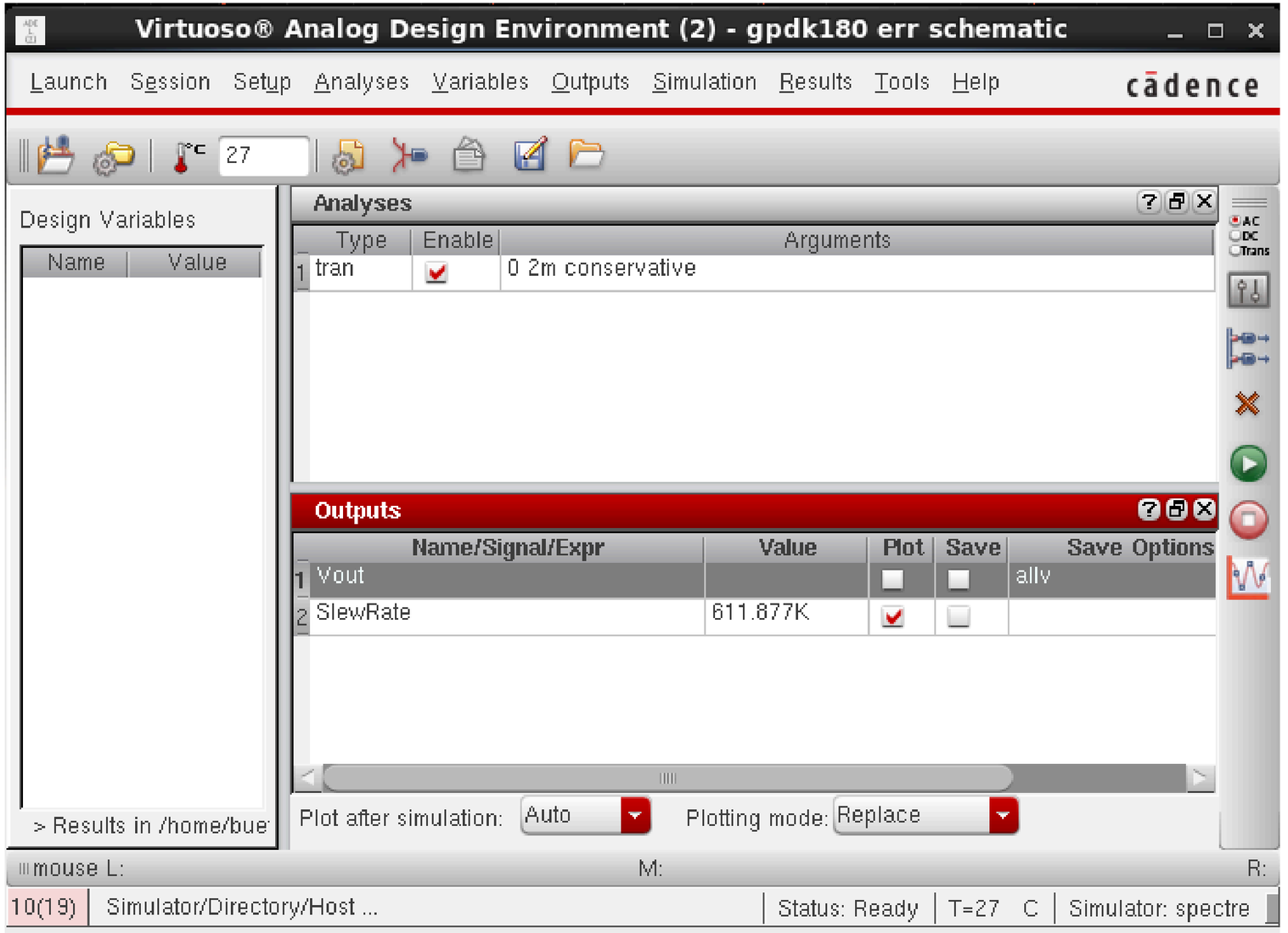
CMRR(Common Mode Rejection Ratio) :



Power :


Virtuoso® Analog Design Environment (1) - gpdk180 err schematic
Launch Session Setup Analyses Variables Outputs Simulation Results Tools Help
cadence
27
Design Variables
Name Value
1 Vcm 0
>
Analyses
Type Enable Arguments
1 dc t 0 1.8 Automatic Start-Stop /Vdd
2 ac 10 1G Automatic Start-Stop
3 tran 0 10m
Outputs
Name/Signal/Expr Value Plot Save Save Options
1 Vout allv
2 Vin- allv
3 Vin+ allv
4 Power 4.7345u
Plot after simulation: Auto
Plotting mode: Replace
AC
DC
Trans
mouse L:
M:
R:

Slew Rate :


Virtuoso® Analog Design Environment (2) - gpdk180 err schematic
Launch Session Setup Analyses Variables Outputs Simulation Results Tools Help
cadence
27
Design Variables
Name Value
> Results in /home/bue
Analyses
Type Enable Arguments
1 tran 0 2m conservative
AC
DC
Trans
Outputs
Name/Signal/Expr Value Plot Save Save Options
1 Vout allv
2 SlewRate 611.877K
Plot after simulation: Auto
Plotting mode: Replace
mouse L: M: R:
10(19) Simulator/Directory/Host ... Status: Ready T=27 C Simulator: spectre

Result :

Parameter / Specification	Target Value	Simulated Value
DC Open-Loop Gain	> 60 – 80 dB	≈ 70.5 dB
Gain Bandwidth Product (UGB)	10 – 2000 kHz	≈ 1629 kHz
Phase Margin	45° - 60°	≈ 51.78°
Supply Voltage	1.0 – 1.8 V	1.8 V
Power Dissipation	< 10 – 100 µW	≈ 4.73 µW
Common-Mode Rejection (CMRR)	> 80 – 100 dB	≈ 87.81
Slew Rate	> 0.1 – 1 V/µs	0.6 V/µs

Conclusion

- Designed and simulated a low-power two-stage error amplifier for an energy-harvesting LDO in 0.18 µm CMOS.
- Selected a PMOS-input differential architecture based on the low input common-mode requirement and used Miller compensation with a nulling resistor for stability.
- Verified the transistor biasing and saturation operation through DC operating-point analysis.
- Achieved approximately 70.5 dB DC open-loop gain with approximately 51.77° phase margin under the simulated conditions.
- Evaluated transient response, bandwidth, unity-gain behavior and CMRR to characterize the amplifier.

Future Work

- Integration of PMU blocks (voltage reference, harvesting interface)
- Line, load, and transient analysis
- PVT and post-layout simulation
- Comparison with state-of-the-art nano-watt PMUs

References

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2. *E. Dallago et al. (2016) – "A 300-mV Low-Power Management System for Energy Harvesting Applications."*
3. *A. A. Blanco and G. A. Rincón-Mora (2018) – "Compact Fast-Waking Light/Heat-Harvesting 0.18-µm CMOS Switched-Inductor Charger."*
4. *M. Seok, G. Kim, D. Blaauw, and D. Sylvester (2012) – "A Portable 2-Transistor Picowatt Temperature-Compensated Voltage Reference Operating at 0.5 V."*
5. *M. D. Seeman and S. R. Sanders (2008) – "Analysis and Optimization of Switched-Capacitor DC–DC Converters."*
6. *H. Lee, S. Kim, and J. Kim (2015) – "A Battery-Less Thermoelectric Energy Harvesting Interface Circuit With 35 mV Startup Voltage."*
7. *Y. K. Ramadass and A. P. Chandrakasan (2010) – "An Efficient Piezoelectric Energy Harvesting Interface Circuit Using a Bias-Flip Rectifier and Shared Inductor."*
8. *H. S. Lee and P. K. Chan (2005) – "A 0.5-V 90-dB Two-Stage CMOS Operational Amplifier With Miller Compensation."*
9. *https://www.researchgate.net/figure/Sources-of-energy-that-can-be-harvested_fig2_351859524*

THANK YOU